

UNIT I: MATRICES, RANK, SYSTEM OF EQUATIONS & EIGENVALUE THEOREMS

Weightage: 25% | Syllabus Section 1

1. Rank, Echelon Form & Normal Form

Rank $\rho(A)$: Number of non-zero rows in row-echelon form. Invariant under elementary row/column transformations.

Normal Form (Canonical Form): Using both row & column operations, reduce A to $[I_r, \ 0; \ 0, \ 0]$. Non-singular matrices P and Q exist such that $P \ A \ Q = [I_r, \ 0; \ 0, \ 0]$, where $\rho(A) = r$.

Gauss-Jordan Inverse: Augment $[A \ | \ I]$. Apply row operations until $[I \ | \ A^{-1}]$ is obtained.

2. Consistency of Linear Systems ($AX = B$)

Rouché-Capelli Theorem: Form augmented matrix $[A \ | \ B]$.

Rank Relation	Consistency	Nature of Solution
$\rho(A) \neq \rho(A B)$	Inconsistent	No Solution
$\rho(A) = \rho(A B) = n$	Consistent	Unique Solution
$\rho(A) = \rho(A B) = r < n$	Consistent	∞ Solutions ($n-r$ free params)

Homogeneous ($AX = 0$): Always consistent (trivial $X=0$). Non-trivial (infinite) solution exists if and only if $|A| = 0$ ($\rho(A) < n$).

3. Eigenvalues & Cayley-Hamilton Theorem

Characteristic Equation: $|A - \lambda I| = 0 \Rightarrow \lambda^3 - S_1 \lambda^2 + S_2 \lambda - |A| = 0$.

- $S_1 = \text{ext}(\text{Trace})(A) = \sum \lambda_i$
- $S_2 = \text{ext}(\text{Sum of minors of diagonal elements})$
- $|A| = \prod \lambda_i$

Properties: Eigenvalues of A^{-1} are $1/\lambda_i$; of A^k are λ_i^k .

Cayley-Hamilton Theorem: Every square matrix satisfies its own characteristic equation: $A^3 - S_1 A^2 + S_2 A - |A| I = 0$.

Application: Multiply by $A^{-1} \Rightarrow A^{-1} = \frac{1}{|A|} [A^2 - S_1 A + S_2 I]$.

UNIT II: DIFFERENTIAL & INTEGRAL CALCULUS, CURVATURE & BETA-GAMMA FUNCTIONS

Weightage: 25% | Syllabus Section 2

1. Rolle's & Mean Value Theorems

Rolle's Theorem: If $f(x)$ is (1) continuous on $[a, b]$, (2) differentiable on (a, b) , (3) $f(a) = f(b)$, then $\exists c \in (a, b)$ such that $f'(c) = 0$.

Lagrange's MVT: Conditions (1) & (2) hold $\Rightarrow \exists c \in (a, b)$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Cauchy's MVT: For $f(x), g(x)$:
$$\left[\frac{f'(c)}{g'(c)} \right] = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Taylor's Series about a:

$$f(x) = f(a) + f'(a)(x-a) + \left[\frac{f''(a)}{2!} \right](x-a)^2 + \dots$$

Maclaurin's Series: Set $a = 0$.

2. Curvature & Radius of Curvature (ρ)

Curvature κ : Rate of turning of tangent: $\kappa = d\psi/ds = \frac{1}{\rho}$.

Form	Formula for Radius of Curvature ρ
Cartesian $y=f(x)$	$\rho = \frac{[1 + (y_1)^2]^{3/2}}{y_2}$
Parametric $x(t), y(t)$	$\rho = \frac{[x'^2 + y'^2]^{3/2}}{ x' y'' - y' x'' }$
Polar $r=f(\theta)$	$\rho = \frac{[r^2 + r_1^2]^{3/2}}{r^2 + 2 r_1^2 - r r_2}$
Pedal $p=f(r)$	$\rho = r \cdot \frac{dr}{dp}$

Centre of Curvature:

$$\bar{x} = x - \frac{y_1(1 + y_1^2)}{y_2}; \bar{y} = y + \frac{(1 + y_1^2)}{y_2} / y_2.$$

3. L'Hospital & Beta-Gamma Functions

L'Hospital's Rule: For $0/0$ or ∞/∞ :

$$\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \lim_{x \rightarrow a} \left[\frac{f'(x)}{g'(x)} \right].$$

Gamma Function: $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx \ (n > 0)$.

• $\Gamma(n+1) = n \Gamma(n) = n!$ (for integer n).

• $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}; \Gamma(n) \Gamma(1-n) = \pi / \sin(n\pi).$

Beta Function: $B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx = \frac{1}{\Gamma(m)\Gamma(n)} \int_0^\pi \sin^{2m-1}\theta \cos^{2n-1}\theta d\theta.$

Fundamental Relation:

$$B(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}.$$

Duplication formula: $\Gamma(m) \Gamma\left(\frac{m+1}{2}\right) = \sqrt{\frac{\pi}{2}} 2^{m-1} \Gamma\left(\frac{2m}{2}\right).$

UNIT III: MULTIVARIABLE CALCULUS, EXTREMA & VECTOR DIFFERENTIAL CALCULUS

Weightage: 25% | Syllabus Section 3

1. Partial Derivatives, Euler's Theorem & Extrema

Euler's Theorem on Homogeneous Functions: If $u(x, y)$ is homogeneous of degree n :
• $x \left(\frac{\partial u}{\partial x} \right) + y \left(\frac{\partial u}{\partial y} \right) = n u$.
• $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = n(n-1) u$.
Total Derivative: $du = \left(\frac{\partial u}{\partial x} \right) dx + \left(\frac{\partial u}{\partial y} \right) dy$.

Discriminant: $\Delta = rt - s^2$	Value of $r = f_{xx}$	Conclusion for Point (a, b)
$rt - s^2 > 0$	$r < 0$	Local Maximum
$rt - s^2 > 0$	$r > 0$	Local Minimum
$rt - s^2 < 0$	Any sign	Saddle Point (Neither max nor min)
$rt - s^2 = 0$	Any	Inconclusive (Further tests required)

Lagrange's Multipliers: Optimize $f(x, y, z)$ subject to $\phi(x, y, z) = 0$. Auxiliary function: $F = f + \lambda \phi \Rightarrow \text{abla} F = 0$.

2. Vector Differential Calculus: $\nabla\phi$, $\nabla\cdot F$, $\nabla\times F$

Del Operator: $\text{abla} = i \left(\frac{\partial}{\partial x} \right) + j \left(\frac{\partial}{\partial y} \right) + k \left(\frac{\partial}{\partial z} \right)$.

Operation	Formula	Physical Meaning & Condition
Gradient ($\nabla\phi$)	$\left(\frac{\partial\phi}{\partial x} \right)i + \left(\frac{\partial\phi}{\partial y} \right)j + \left(\frac{\partial\phi}{\partial z} \right)k$	Normal vector to surface $\phi = c$. Max rate of increase = $ \text{abla}\phi $.
Directional Derivative	$D_a\phi = \text{abla}\phi \cdot \frac{a}{ a }$	Rate of change of ϕ along direction vector a .
Divergence ($\nabla\cdot F$)	$\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$	Net outward flux per volume. $\text{abla} \cdot F = 0 \Rightarrow$ Solenoidal Field .
Curl ($\nabla\times F$)	Determinant with $[i, j, k; \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}; F_1, F_2, F_3]$	Rotational vorticity. $\text{abla} \times F = 0 \Rightarrow$ Irrotational / Conservative Field ($F = \text{abla}\phi$).

Fundamental Vector Null Identities:

- $\text{abla} \times (\text{abla}\phi) = 0$ (Curl of any gradient is identically zero).
- $\text{abla} \cdot (\text{abla} \times F) = 0$ (Divergence of any curl is identically zero).

UNIT IV: CONVERGENCE TESTS OF SEQUENCES & SERIES AND FOURIER EXPANSIONS

Weightage: 25% | Syllabus Section 4

1. Infinite Series Convergence Tests

Test Name	Test Limit Ratio: L	Convergence Criterion
D'Alembert Ratio Test	$\lim_{n \rightarrow \infty} \left \frac{u_n}{u_{n+1}} \right = L$	$L > 1 \Rightarrow \text{ext(Conv)}; L < 1 \Rightarrow \text{ext(Div)}; L = 1 \Rightarrow \text{ext(Fails)}$
Raabe's Test (Use when Ratio fails)	$\lim_{n \rightarrow \infty} n \left[\left(\frac{u_n}{u_{n+1}} \right) - 1 \right] = L$	$L > 1 \Rightarrow \text{ext(Conv)}; L < 1 \Rightarrow \text{ext(Div)}; L = 1 \Rightarrow \text{ext(Fails)}$
Logarithmic Test	$\lim_{n \rightarrow \infty} \left[n \ln \left(\frac{u_n}{u_{n+1}} \right) \right] = L$	$L > 1 \Rightarrow \text{ext(Conv)}; L < 1 \Rightarrow \text{ext(Div)}$
Cauchy's Root Test	$\lim_{n \rightarrow \infty} (u_n)^{1/n} = L$	$L < 1 \Rightarrow \text{ext(Conv)}; L > 1 \Rightarrow \text{ext(Div)}$
Leibnitz Rule (Alternating)	$\sum (-1)^{n-1} u_n$	Converges if (1) $u_{n+1} \leq u_n$, (2) $\lim u_n = 0$

- Absolute Convergence:** $\sum |u_n|$ converges.
- Conditional Convergence:** $\sum u_n$ converges, but $\sum |u_n|$ diverges.

2. Fourier Series & Half-Range Expansions

Dirichlet's Conditions: $f(x)$ has Fourier expansion if in interval: (1) single-valued & bounded, (2) finite number of points of discontinuity, (3) finite number of maxima and minima.

Fourier Expansion in $[-L, L]$:

$f(x) = \left(\frac{a_0}{2} \right) + \sum_{n=1}^{\infty} \left[a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right]$

- $a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$
- $a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx$
- $b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx$

Even vs. Odd Functions in $[-L, L]$:

- Even $[f(-x) = f(x)]$:** $b_n = 0 \Rightarrow$ Fourier Cosine Series: $a_0 = \frac{2}{L} \int_0^L f dx, a_n = \frac{2}{L} \int_0^L f \cos \frac{n\pi x}{L} dx$.

- Odd $[f(-x) = -f(x)]$:** $a_0 = 0, a_n = 0 \Rightarrow$ Fourier Sine Series: $b_n = \frac{2}{L} \int_0^L f \sin \frac{n\pi x}{L} dx$.

Half-Range Expansions in $[0, L]$: Extended as even (Cosine) or odd (Sine).